

Mitschrift

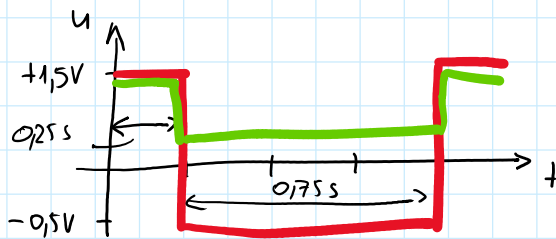
Samstag, 18. Jänner 2025 08:53

Inhalt:

- Formelsammlung
- Zeitabhängigkeit Bsp. E9
- Zusammenhang elektrische Größen
- WH lineare Zweipole an Wechselspannung
- Bsp. 5.2
- Ltspice
- Bsp. Python
- Leistung in Wechselstromtechnik
- Resonanz
- Bsp. 5.15
- Drehstromtechnik
- Bsp. 7.2
- Stromversorgung

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ZEITABHÄNGIGKEIT

periodisches Signal

$$T = 1s \quad f = 1Hz$$

$$\omega = 2\pi f = 2\pi s^{-1}$$

$$U_{\max} = +1,5V \quad U_{\min} = -0,5V \quad \hat{U} = 1,5V \quad U_{pp} = 2V$$

$$\bar{U} = \frac{1}{T} \int_0^T u \, dt = \frac{1}{1s} \left(\int_0^{0,25s} 1,5V \, dt + \int_{0,25}^{1,0} -0,5 \, dt \right) =$$

$$= \frac{1}{1s} \left(\left[1,5t \right]_0^{0,25} + \left[-0,5t \right]_{0,25}^{1,0} \right) = \frac{1}{1s} (0,375 - 0 - 0,5 + 0,125) Vs =$$

$$= \frac{1}{1s} 0Vs = \underline{0V}$$

$$|\bar{U}| = \frac{1}{T} \int_0^T |u| \, dt = \frac{1}{1s} \left(\int_0^{0,25} 1,5V \, dt + \int_{0,25}^{1,0} 0,5V \, dt \right) =$$

$$= \frac{1}{1s} (0,375 - 0 + 0,5 - 0,125) Vs = 0,75V$$

$$U = \sqrt{\frac{1}{T} \int_0^T u^2 \, dt}$$

$$\int_0^T u^2 \, dt = \int_0^{0,25} 1,5^2 V^2 \, dt + \int_{0,25}^{1,0} 0,5^2 V^2 \, dt =$$

$$= 2,25V^2 t \Big|_0^{0,25} + 0,25V^2 t \Big|_{0,25}^{1,0} =$$

$$= (2,25 \cdot 0,25 - 0 + 0,25 \cdot 1 - 0,0625) V^2 s$$

$$= 0,75V^2 s$$

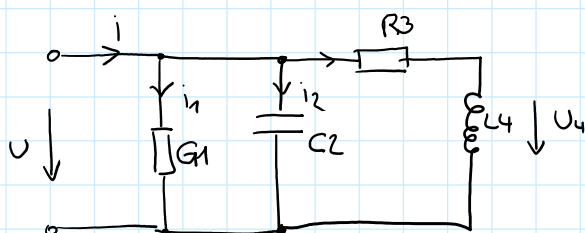
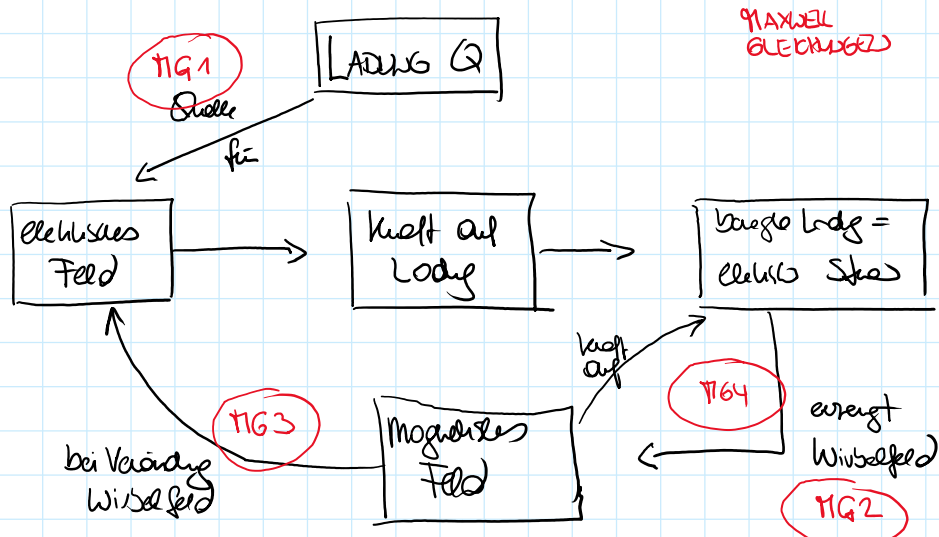
$$\underline{U} = \sqrt{\frac{1}{1s} 0,75V^2 s} = \underline{0,866V}$$

$$\text{Sollwert} = \hat{U} = \underline{1,5V} = 1,732$$

$$\text{Scheitelfaktor} = \frac{\hat{u}}{U} = \frac{1,5V}{0,866V} = 1,732$$

$$\text{Formfaktor} = \frac{U}{\overline{u}} = \frac{0,866V}{0,775V} = 1,114$$

WECHSELSTROMTECHNIK



$$u = 5V \cdot \cos(6283 s^{-1} t)$$

$$G_1 = 1mS \quad C_2 = 120nF \quad R_3 = 1,5k\Omega \quad L_4 = 159mH$$

a) $\omega = 2\pi f = 6283 s^{-1} \rightarrow f = 1kHz$

$$\hat{u} = 5V \quad \varphi_u = 0^\circ \quad U = \frac{5}{\sqrt{2}} V = 3,54V$$

$$\underline{U} = 3,54V \angle 0^\circ$$

b) $\underline{Y} \quad \underline{Z}$

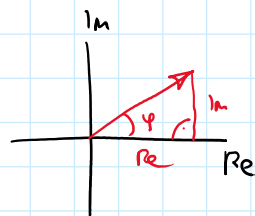
$$\underline{Z}_{34} = R_3 + j\omega L_4 = 1500\Omega + j999\Omega$$

$$\underline{Y}_{34} = \frac{1}{\underline{Z}_{34}} = \frac{1}{1500\Omega + j999\Omega} = \frac{1500 - j999}{1500^2 + 999^2} \frac{1}{\Omega} =$$

$$= 0,461mS - j0,308mS = 0,57mS \angle -33,7^\circ$$

$$\underline{Y} = G_1 + j\omega C_2 + \underline{Y}_{34} = 1mS + j0,754mS + 0,461mS - j0,308mS =$$

$$= 1,461mS + j0,446mS = 1,528mS \angle 17^\circ$$



$$\underline{Z} = \frac{1}{\underline{Y}} = 654,6 \, \Omega \angle -17^\circ$$

$$c) \quad \underline{I}_2 = \frac{\underline{U}}{\underline{Z}_{C2}} = j\omega C_2 \cdot \underline{U} = 0,754 \text{ mS} \angle 90^\circ \cdot 3,54 \text{ V} \angle 0^\circ = 2,67 \text{ mA} \angle 90^\circ$$

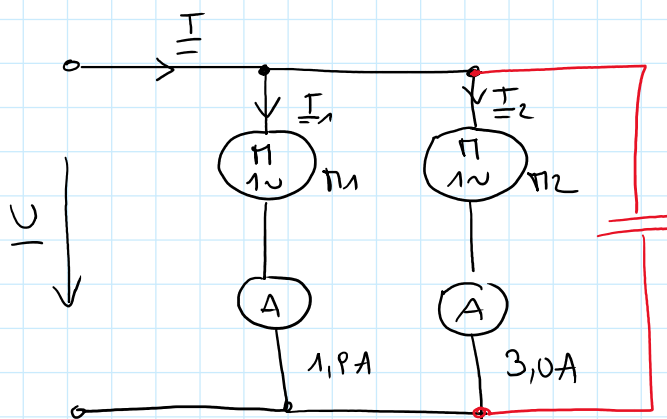
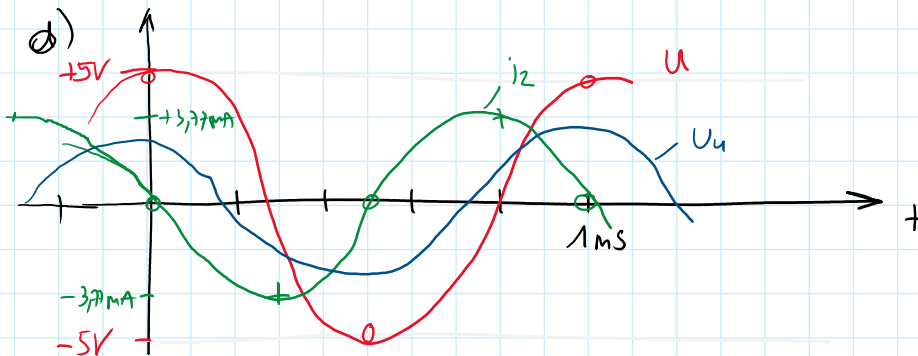
$$i_2(t) = 3,77 \text{ mA} \cos(6283 \, s^{-1} t + 90^\circ)$$

$$\underline{U}_4 = \frac{j\omega L_4}{\underline{Z}_{34}} \underline{U} = \frac{j 999 \, \Omega}{1500 \, \Omega + j 999 \, \Omega} 3,54 \text{ V} \angle 0^\circ =$$

$$= (0,307 + j 0,461) \cdot 3,54 \text{ V} =$$

$$= 1,087 \text{ V} + j 1,632 \text{ V} = 1,961 \text{ V} \angle 56,3^\circ$$

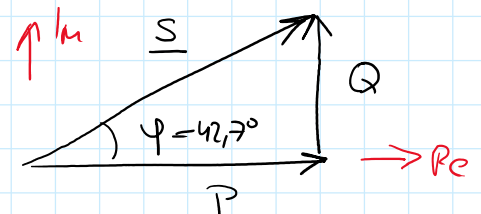
$$u_4 = 2,773 \text{ V} \cos(6283 \, s^{-1} t + 56,3^\circ)$$



$$U = 230 \text{ V} \quad f = 50 \text{ Hz}$$

$$P_R = 400 \text{ W}$$

R_1 teilkompensiert



a) bestimme $\underline{I}$ sowie P, Q, S

$$S_1 = U \cdot I_1 = 437 \text{ VA}$$

$$S_2 = U \cdot I_2 = 690 \text{ VA}$$

$$\cos \varphi_1 = \frac{P}{S_1} = \frac{400}{437} = 0,915$$

$$\cos \varphi_2 = \frac{P_2}{S_2} = \frac{400}{690} = 0,58$$

$$\varphi_1 = 23,8^\circ$$

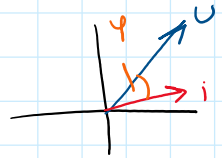
$$\varphi_2 = 54,6^\circ$$

14 $\nearrow U$

$$\varphi_1 = 23,8^\circ$$

$$\varphi_2 = 54,6^\circ$$

$$\varphi = \varphi_u - \varphi_i$$



Annahme $\underline{U} = 230V \angle 0^\circ$

$$\varphi_{i1} = -23,8^\circ$$

$$\varphi_{i2} = -54,6^\circ$$

$$\underline{I}_1 = 1,9A \angle -23,8^\circ$$

$$\underline{I}_2 = 3,0A \angle -54,6^\circ$$

$$\underline{I} = \underline{I}_1 + \underline{I}_2 = 4,73A \angle -42,7^\circ$$

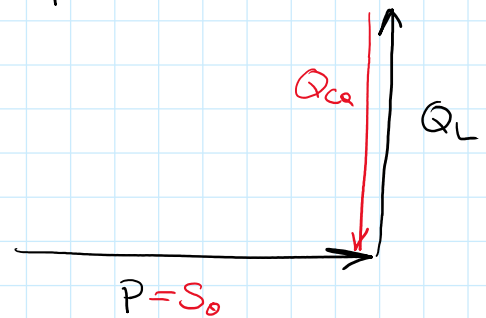
$$\underline{S} = \underline{U} \underline{I}^* = 230V \angle 0^\circ \cdot 4,73A \angle 42,7^\circ = 1.088 VA \angle 42,7^\circ =$$

$$= 800W + j 738 var = P + j Q$$

b) kompensieren $\cos \varphi_a = 1$ $\cos \varphi_s = 0,95$

$$Q_{co} = -Q_L = -738 var = -\omega C_o U^2$$

$$C_o = -\frac{Q_{co}}{\omega U^2} = 44,4 \mu F$$



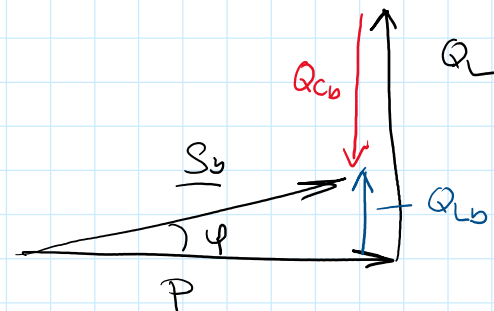
$$\cos \varphi_b = 0,95 \quad + \quad P = \text{konst.}$$

$$\varphi_b = 18,2^\circ$$

$$Q_L = P \cdot \tan \varphi_b = 263 var$$

$$Q_{cb} = -(Q_L - Q_{Lb}) = -475 var$$

$$C_b = -\frac{Q_{cb}}{\omega U^2} = 28,6 \mu F$$



2025.01.17 GLET1 / Gefahren elektrischen Stroms

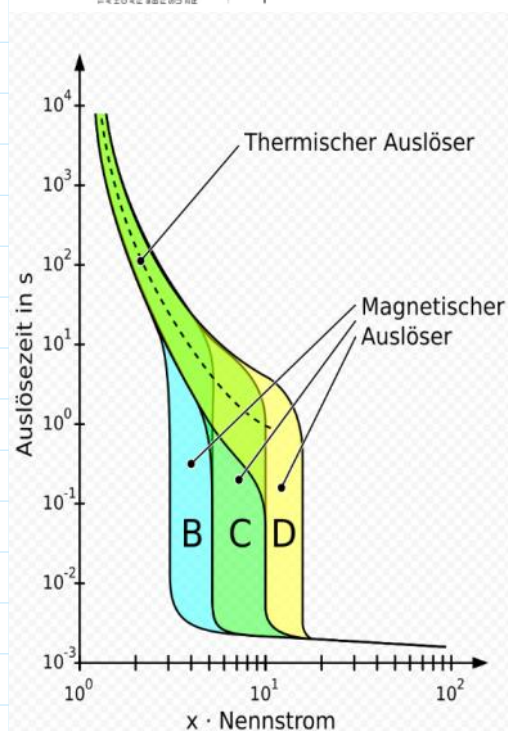
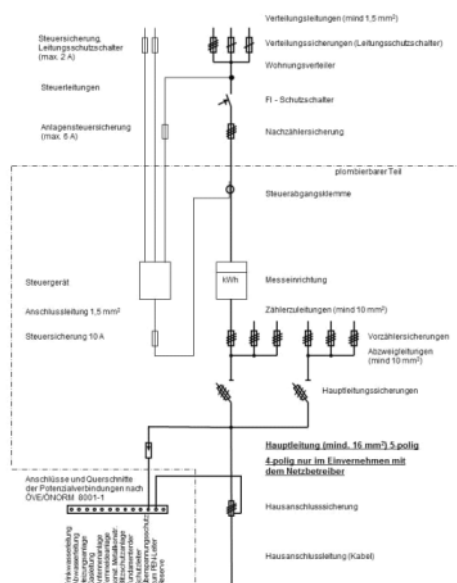
Donnerstag, 16. Jänner 2025 20:18

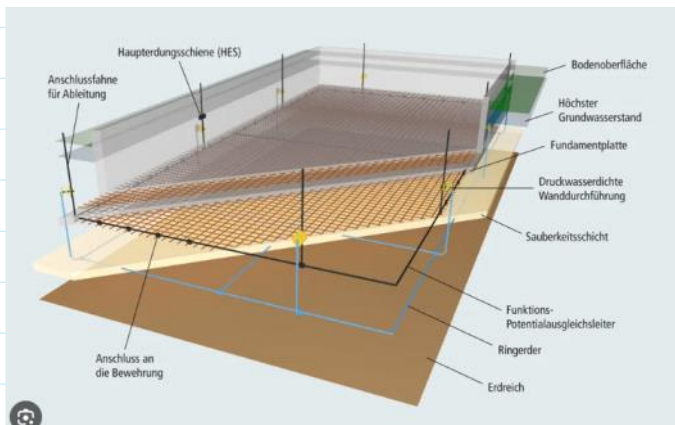
Lichtbogen 5000 A

[Kurzschluss Lichtbogen, 400 V, 1.2 Sekunden, 5000 Ampere](#)

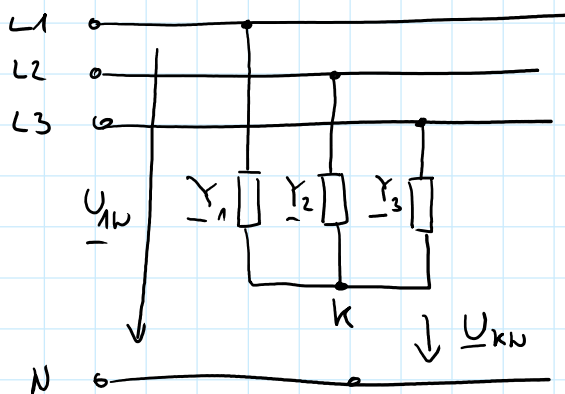


Abb. 1: Begriffsbestimmungen bei der Hausinstallation





BSP DREHSTROM



$$U_{\Delta} = 400V$$

$$f = 50Hz$$

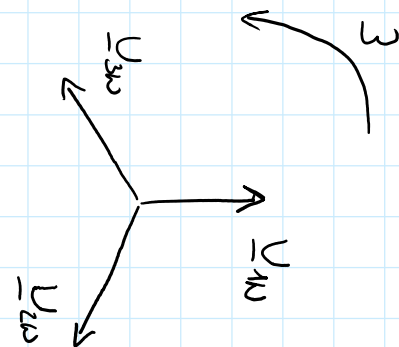
$$\underline{Y}_1 = 10mS \angle 0^\circ$$

$$\underline{Y}_2 = 20mS \angle 60^\circ$$

$$\underline{U}_{KN} = \frac{\underline{Y}_1 \cdot \underline{U}_{1N} + \underline{Y}_2 \cdot \underline{U}_{2N} + \underline{Y}_3 \cdot \underline{U}_{3N}}{\underline{Y}_1 + \underline{Y}_2 + \underline{Y}_3}$$

Symmetrisches Spannungssystem $\rightarrow \underline{U}_{KN} = 0$

$$\underline{Y}_3 = - \frac{\underline{Y}_1 \cdot \underline{U}_{1N} + \underline{Y}_2 \cdot \underline{U}_{2N}}{\underline{U}_{3N}}$$

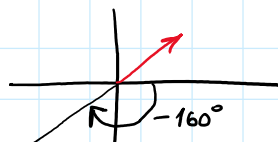


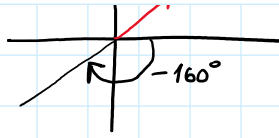
$$\underline{U}_{1N} = U_{\Delta} \angle 0^\circ \quad \underline{U}_{2N} = U_{\Delta} \angle -120^\circ \quad \underline{U}_{3N} = U_{\Delta} \angle +120^\circ$$

$$\underline{Y}_3 = - \frac{10mS \angle 0^\circ + 20mS \angle -60^\circ}{1 \angle +120^\circ} = \frac{10mS + 10mS - j17,3mS}{1 \angle +120^\circ} =$$

$$= - \frac{20mS - j17,3mS}{1 \angle +120^\circ} = - \frac{26,4mS \angle -40,9^\circ}{1 \angle +120^\circ} = -26,4mS \angle 160,9^\circ$$

$$= 26,4mS \angle 19,1^\circ$$

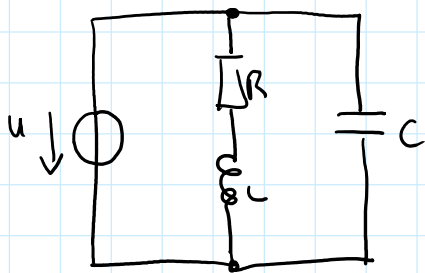
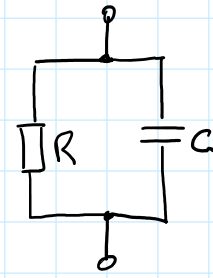




$$\underline{Y}_3 = G + jB = (25 + j 8,6) \text{ mS}$$

$$R = 1/G = 40 \Omega$$

$$B = \omega C \rightarrow C = \frac{B}{\omega} = \frac{8,6 \text{ mS}}{2\pi f} = 27,6 \mu\text{F}$$



$$R = 1 \Omega \quad L = 10 \mu\text{H}$$

$$\underline{U} = 20 \text{ V} \angle 0^\circ \quad f = 80 \text{ kHz}$$

ges. C bei $f_{\text{res}} = 80 \text{ kHz}$

$$\frac{Q+B}{C} = \frac{Q}{C} + \frac{B}{C}$$

$$\underline{Y} = j\omega C + \frac{1}{R + j\omega L} = j\omega C \cdot \frac{R - j\omega L}{R^2 + \omega^2 L^2}$$

$$B = \omega C - \frac{\omega L}{R^2 + \omega^2 L^2} = 0$$

$$C = \frac{L}{R^2 + \omega^2 L^2} =$$

$$\frac{R}{R^2 + \omega^2 L^2} - \frac{j\omega L}{R^2 + \omega^2 L^2}$$

$$\underline{Y} = G + jB = 380 \text{ nF}$$

$$\underline{Y} = G = \frac{R}{R^2 + \omega^2 L^2} = 38 \text{ mS}$$

$$R_e = \frac{1}{G} = 26,3 \Omega$$

$$\underline{I} = G \cdot \underline{U} \quad \underline{I} = 760 \text{ mA}$$